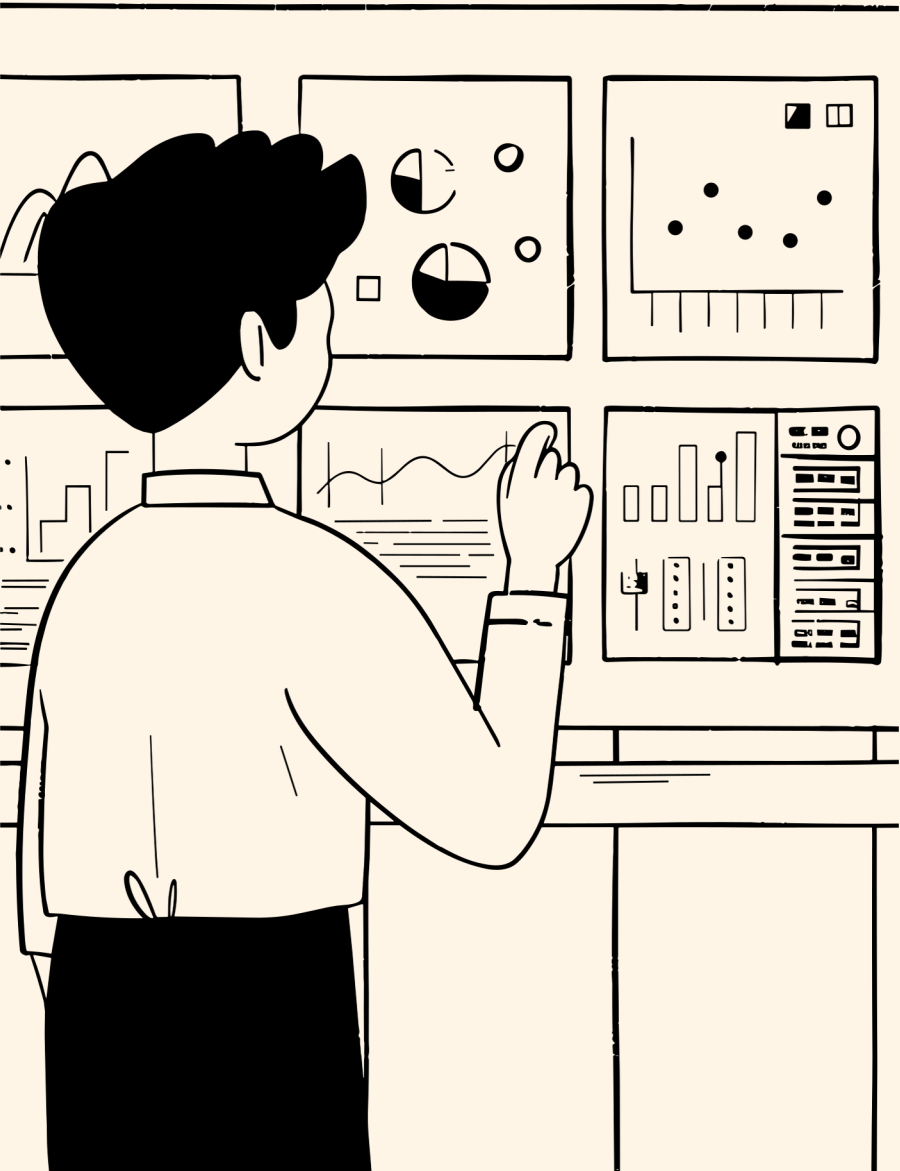




Customer Shopping Behavior Analysis

Using Python, PostgreSQL (Neon), SQL and Power BI—by Anurag Mishra. A 3,900-record study that turns raw transactions into actionable business insights.

Pipeline Overview

Data

customer_shopping_behavior.csv — 3,900 records

Processing

Python & Pandas:
cleaning, missing-value handling, feature engineering

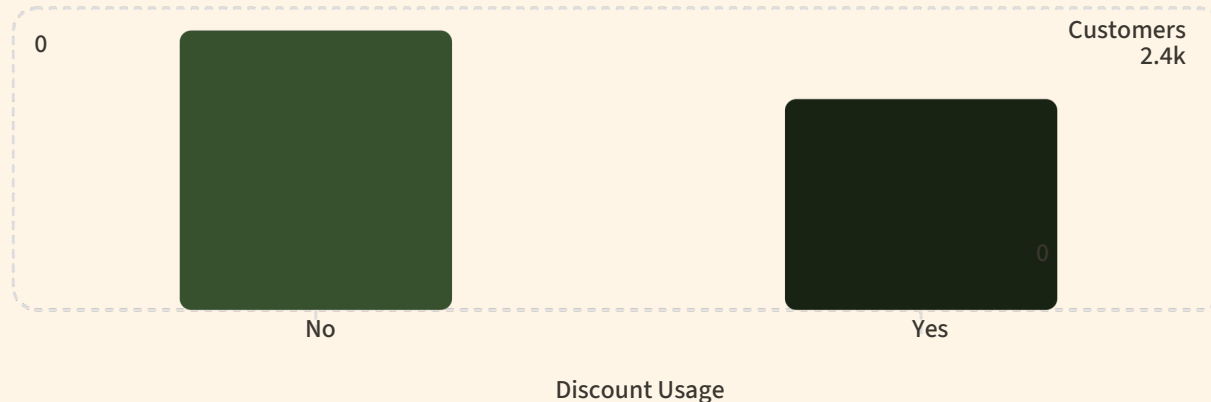
Storage & Analysis

PostgreSQL on Neon —
SQL queries for business questions

Visualization

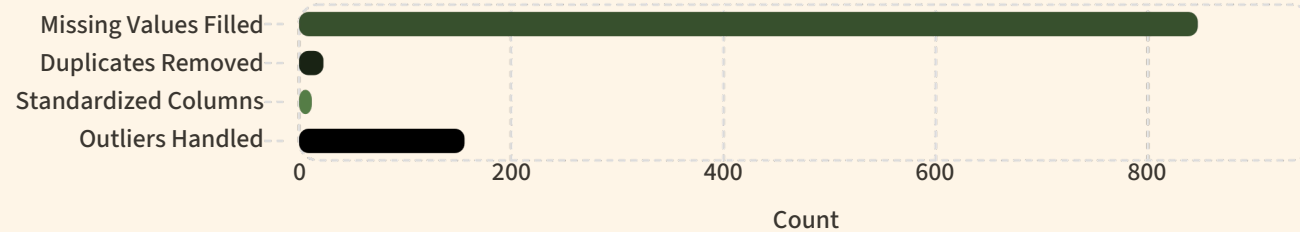
Power BI dashboard
with DAX KPIs and interactive filters

Discount Usage



Preprocessing Highlights

Data Quali...



Missing Ratings

Filled missing review_rating using category-wise median (median chosen to reduce outlier influence).

Column Standardization

Lowercased names, replaced spaces with underscores, renamed purchase_amount column for SQL compatibility.

Redundancy Removal

Promo_code_used was dropped after confirming it duplicated discount_applied.

Frequency → Days

Mapped categorical frequency to days (Weekly=7, Monthly=30, Quarterly=90, etc.).

Feature Engineering

Age Groups

Quartile-based bins: Young Adult, Adult, Middle-aged, Senior — used for revenue analysis.

Purchase Frequency Days

Categorical frequencies converted to numeric days for behavioral analysis.

Derived Columns

Standardized column names and removed redundant promo_code flag to reduce noise.



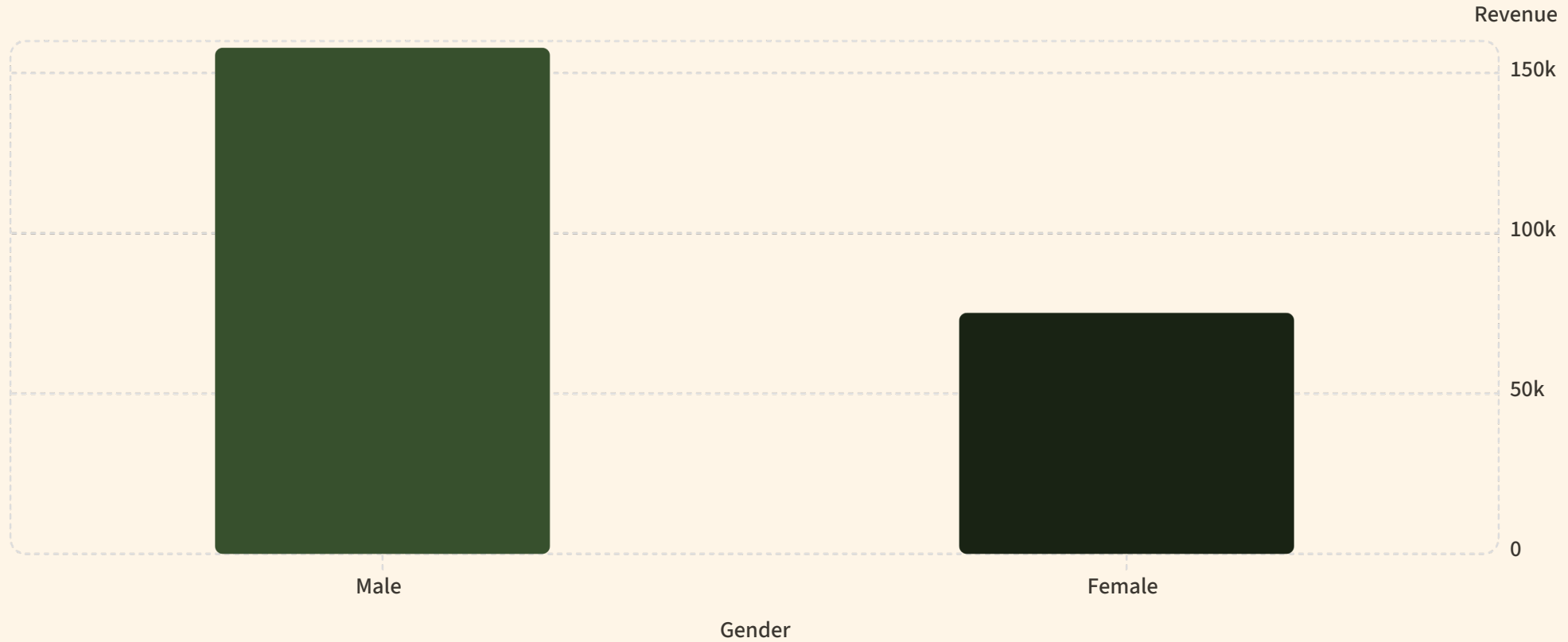


PostgreSQL (Neon) Integration

CleanedDataFrameuploadedtoaNeon-hostedPostgreSQL table named "customer" via `df.to_sql`. Connection verified with `SELECT version()`; ready for SQL analysis.

Key SQL Findings (Revenue & Discounts)

Revenue



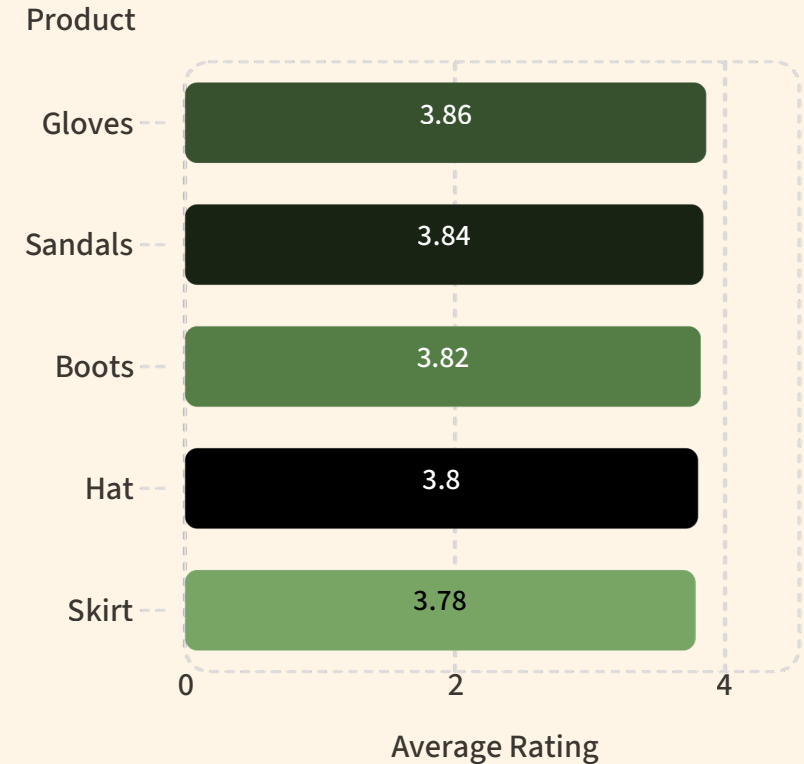
Product & Category Insights

Top Rated Products

Top 5 by average rating: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78).

Top Items by Orders

Per category examples: Accessories — Jewelry, Sunglasses, Belt. Clothing — Blouse, Pants, Shirt. Footwear — Sandals, Shoes, Sneakers.



Customer Behavior & Segmentation

Shipping vs Spend

Express avg purchase: 60.48 — Standard avg purchase: 58.46 (difference 2.02).

Association noted, not causation.

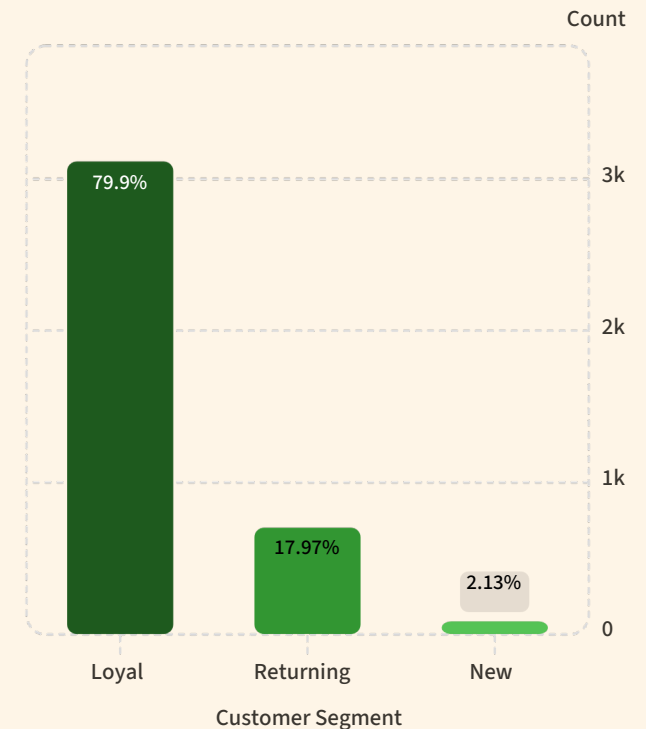
Subscription Analysis

Subscribers: 1,053 (avg spend 59.49, total 62,645). Non-subscribers: 2,847 (avg 59.87, total 170,436).

Customer Segments

Defined by previous_purchases: New (≤ 1), Returning (2–10), Loyal (>10).

Counts — Loyal: 3,116; Returning: 701; New: 83.



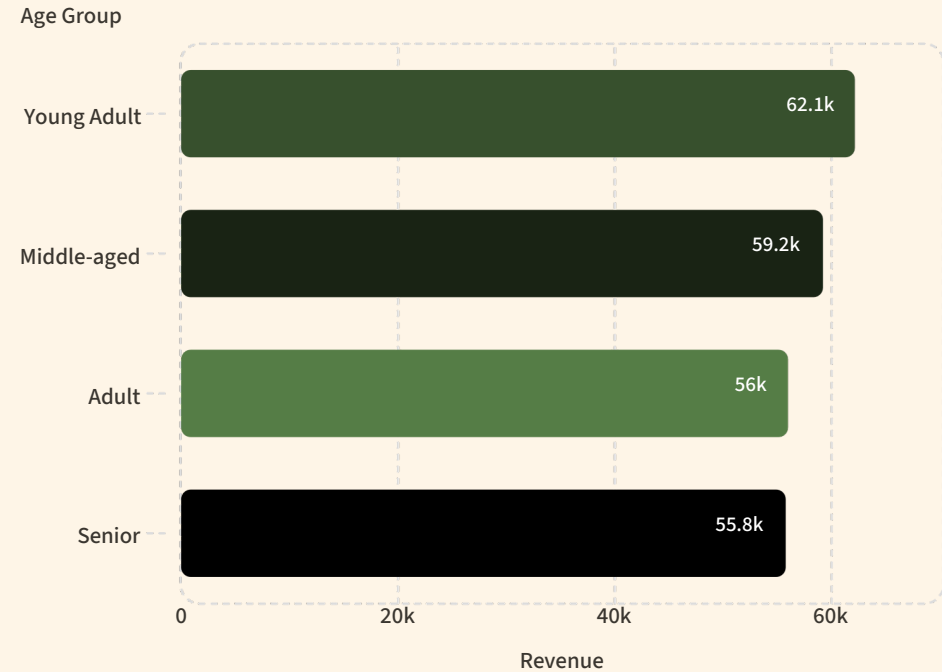
Repeat Buyers & Age-Group Revenue

Repeat Buyers

Repeat buyers (previous_purchases > 5): 3,476 total — Non-subscriber repeat buyers: 2,518; Subscriber repeat buyers: 958.
Opportunity to convert repeat buyers to subscribers.

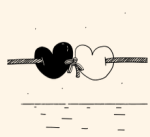
Revenue by Age Group

Young Adult: 62,143 (highest) • Middle-aged: 59,197 • Adult: 55,978 • Senior: 55,763.



Revenue is highest among Young Adults, followed closely by Middle-aged customers.

Conclusions & Business Recommendations



Reward Loyalty

Introduce loyalty points, exclusive offers and early access — loyal customers (3,116) are a priority for retention.



Improve Subscription Conversion

Target repeat non-subscribers with incentives (discounts, free shipping) to raise subscription uptake.



Target Young Adults

Young Adults generate the most revenue (62,143). Tailor campaigns and product placement toward this group.



Optimize Discounts

Monitor high-discount items (Hat 50%) to ensure discounts drive profitable volume and not margin erosion.

Limitations: single dataset scope, no causal claims, external factors excluded. Future scope: churn prediction, recommendations, CLV, advanced segmentation, real-time dashboards.